

GREEN ENLIGHTENMENT

AI & Sentinel-2 Satellite Multi-Spectral Agroforestry MRV Platform
Official Grant & Seed Funding Budget Proposal (ACIC / NITI Aayog)

FUNDING ASK: INR 25,00,000
12-Month Deployment Plan

1. EXECUTIVE SUMMARY & TARGET OBJECTIVES

Green Enlightenment solves India's 60%+ sapling mortality crisis and corporate greenwashing by deploying an automated AI-Vision + Sentinel-2 Satellite MRV (Measurement, Reporting, and Verification) platform. This 12-month budget proposal represents an honest, capital-efficient, and defensible financial roadmap tailored for the Atal Community Innovation Centre (ACIC) to scale grassroots agroforestry in rural Tier-2/3 India.

2. 12-MONTH MACRO CAPITAL ALLOCATION SUMMARY

Budget Category	Amount (INR)	% Share	Primary Deliverable / Scope
1. Product Engineering, AI & Satellite Cloud	INR 7,20,000	28.8%	Sentinel-2 pipeline, Gemini 2.5 compute, Supabase, offline PWA
2. Grassroots Field Pilots & Rural Deployment	INR 6,00,000	24.0%	20k physical QR tags, 5 GPS scouting kits, farmer bio-inputs
3. Core Technical & Agroforestry Talent	INR 7,80,000	31.2%	Lead AI/GIS Engineer, Agroforestry Lead, Rural Field Coordinator
4. Carbon MRV Audits, IPCC Compliance & IP	INR 2,20,000	8.8%	IPCC Tier-2 scientific validation, provisional patent filing, legal
5. Vernacular Outreach, Workshops & Buffer	INR 1,80,000	7.2%	10 Marathi/Hindi KVK workshops, printed manuals, 7% reserve
TOTAL 12-MONTH BUDGET	INR 25,00,000	100.0%	Milestone Target: 50,000 verified trees & 2 paid corporate pilots

3. DETAILED LINE-ITEM BREAKDOWN (PART A)

Category 1: Technology & AI Infrastructure	Technical Description	Cost (INR)
Copernicus Sentinel-2 & Planet Tile Pipeline	Cloud storage, GeoTIFF tiling, and spectral processing for 100k+ trees	INR 2,40,000
Google Gemini 2.5 Vision API Inference	Species verification, plant pathology & anti-fraud auto-rejection (~50k calls)	INR 1,80,000
Supabase Enterprise Vector & Cloud Database	PostgreSQL spatial DB, encrypted image vaults, real-time edge functions	INR 1,20,000
Offline PWA Engine & Vector QR Tag Pipeline	Offline IndexedDB synchronization & raster-to-SVG batch print exporter	INR 1,00,000
CI/CD, High-Availability Hosting & DNS	Vercel Enterprise, edge caching, serverless workers, SSL security	INR 80,000
CATEGORY TOTAL: TECH & AI INFRASTRUCTURE	28.8% of Total Capital	INR 7,20,000

Category 2 to 5: Field, Talent, Carbon & Outreach		Operational Scope	Cost (INR)
Physical Weatherproof QR Branch Tags	20,000 UV-resistant anodized aluminum QR tree tags @ INR 12/tag		INR 2,40,000
Field Scouting Kits for Rural Rangers	5 rugged GPS/NFC field tablets + optical soil moisture probes		INR 1,50,000
Pilot Land Parcels & Farmer Inputs	Subsidized bio-fertilizer kits (Trichoderma, Jeevamrit) for 500 farmers		INR 1,50,000
Field Logistics & Regional Travel	Travel across Solapur, Satara, Ahmednagar & Vidarbha agroforestry clusters		INR 60,000
Lead AI/Geospatial Full-Stack Engineer	12 Months lean incubation stipend @ INR 30,000 / month		INR 3,60,000
Agroforestry & Plant Pathology Lead	Part-time domain specialist @ INR 15,000 / month		INR 1,80,000
Rural Field Coordinator & Community Lead	12 Months on-ground liaison @ INR 20,000 / month		INR 2,40,000
IPCC Tier-2 Pantropical Model Audit	Independent university forestry department validation & certification		INR 1,00,000
Patent / IP Filing & Legal Compliance	Provisional patent on AI Anti-Fraud Geo-Spatial Algorithm + GST/Audit		INR 1,20,000
Vernacular Farmer Training & Buffer	10 Marathi/Hindi KVK workshops, printed guides + 7% contingency reserve		INR 1,80,000
TOTAL BUDGET BALANCE	Full 12-Month Operational & Technical Runway		INR 17,80,000

4. MONETIZATION & COMMERCIAL SELF-SUSTAINABILITY (NO PERPETUAL GRANTS)

- Corporate ESG MRV Audits: INR 15 - INR 25 per tree/year for automated multi-spectral satellite verification.
- Bulk Weatherproof QR Tag Bundles: INR 20 per physical anodized aluminum tag bundle for mass plantation drives.
- Carbon Offset Issuance Brokerage: 8% - 12% brokerage fee on verified carbon credits under IPCC Tier-2 standards.
- Breakeven Runway: At 1,00,000 monitored trees, annual recurring revenue reaches INR 18,00,000, achieving full self-sufficiency.

5. 12-MONTH TARGET KEY PERFORMANCE INDICATORS (KPIs)

Key Impact Metric	Target by Month 12	Strategic Value to ACIC / NITI Aayog
Trees Geotagged & Audited	50,000+ Living Trees	Compared to standard 40% unmonitored baseline
24-Month Sapling Survival Rate	>90% Verified Survival	Powered by IoT/satellite soil moisture & AI scouting alerts
CO2 Equivalent Sequestered	1,00,000+ kg CO2e (1,000 MT)	Quantified under IPCC Pantropical Tier-2 allometric models
Corporate CSR Pilots	2 Paid Enterprise Clients	Direct pathway to early commercial revenue by Month 9